

Program 4 : Integrate Git with Docker for source controlled application builds.

Step 1: make a folder -> program-4 and move into the folder

```
mkdir program-4
```

```
cd program-4
```

Step 2: Create the file -> package.json by executing code

```
npm init -y
```

Step 3: Create [index.js](#)

```
GNU nano 7.2 index.js
const express=require("express");
const app=express();
const PORT = 3000;
app.get("/",(req,res)=>{
    res.send("Hello from Git + Docker Integration!");
});
app.listen(PORT,()=>{
    console.log('Server running at http://localhost:${PORT}');
});
```

Step 4: Add dependency

```
npm install express
```

Step 5: Create Dockerfile

```
GNU nano 7.2 Dockerfile
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY ..

FROM node :18-slim
WORKDIR /app
COPY --from=build /app/app
EXPOSE 3000
CMD["node","index.js"]
```

Step 6: Create new Repo on Github

Step 7: Get Personal Access Token

Go to profile-> Developer Setting -> Personal Access Token -> Token(Classic) and generate a Token.

The screenshot shows the GitHub interface for creating a new Personal Access Token (Classic). On the left, there are tabs for 'Fine-grained tokens' and 'Tokens (classic)'. The main form includes the following sections:

- Token name ***: A text input field containing 'program'. A green checkmark indicates 'program' is available. A note states: 'A unique name for this token. May be visible to resource owners or users with possession of the token.'
- Description**: A large text area for describing the token's purpose.
- Resource owner**: A dropdown menu showing 'Akash3639'. A note states: 'The token will only be able to make changes to resources owned by the selected resource owner. Tokens can always read all public repositories.'
- Expiration**: A date picker set to '30 days (Dec 21, 2025)'. A note states: 'The token will expire on the selected date.'
- Repository access**: Three radio button options: 'Public repositories' (Read-only access to public repositories), 'All repositories' (This applies to all current and future repositories you own. Also includes public repositories (read-only)), and 'Only select repositories' (Selected). The 'Only select repositories' option is selected. Below it, a 'Select repositories' button shows '1' repository selected, listed as 'Akash3639/program-4'.
- Permissions**: A section titled 'Choose the minimal permissions necessary for your needs. [Learn more about permissions.](#)'. It contains a table with two rows: 'Contents' (Repository contents, commits, branches, downloads, releases, and merges. [Learn more.](#)) with 'Access: Read and write' and 'Metadata' (Search repositories, list collaborators, and access repository metadata. [Learn more.](#)) with 'Access: Read-only'. A 'Required' badge is next to the Metadata row. An 'Add permissions' button is at the top right of the table.

Step 8: Initialize Git Repository locally.

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
```

Step 9: Link to remote Repository

```
git remote add origin http://github.com/<username>/<repo>.git
```

Step 10: Push code to Github

```
git push -u origin main
```

Akash3639 / program-4

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About

Akash3639

Initial commit

425cc5c · 26 minutes ago · 1 Commit

node_modules	Initial commit	26 minutes ago
Dockerfile	Initial commit	26 minutes ago
index.js	Initial commit	26 minutes ago
package-lock.json	Initial commit	26 minutes ago
package.json	Initial commit	26 minutes ago

No description, website, or topics provided.

Activity

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Releases

No releases published

Create a new release